STP-009 / RESEARCH WORKING PAPER

The Nullspace Has No Name

Identifiability, Source Attribution, and Refusal in Multi-Machine Vibration Monitoring

Abstract

When several rotating machines excite the same structure, a vibration sensor measures a mixture rather than a machine. Blind source separation, spectral features, and even an apparently perfect multichannel reconstruction can produce plausible components without proving which physical motor generated them. This paper formalizes that gap using a frequency-domain transfer model and two elementary identifiability results: a square full-rank model can fit every observed mixture while hiding an unmodeled source, and residual-based detection is blind to interference lying inside the calibrated transfer subspace. It proposes a Source Attribution Evidence Contract (SAEC) that binds each attribution claim to sensor identity, timing quality, transfer-path calibration, numerical conditioning, independent source references, counterfactual tests, and an explicit right to abstain. The contract is examined as a research proposal, not a deployed capability, against blind separation, order tracking, nonnegative factorization, and supervised diagnosis. A falsifiable three-motor/fourth-source protocol specifies the evidence needed to move from synthetic reconstruction to defensible industrial attribution. SynapticSteel dual-node engineering experiments motivate the problem but do not constitute field validation.

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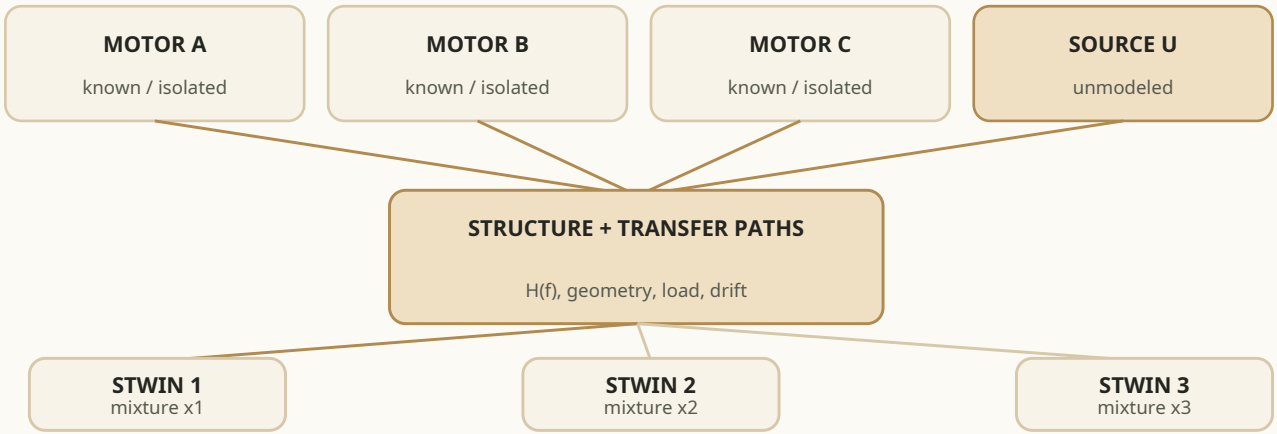
1. The machine behind the measurement

A crowded machine room defeats the convenient assumption that a sensor belongs to the nearest machine. A motor excites foundations, piping, rack structures, housings and air; those paths couple vibration and acoustic energy into sensors attached elsewhere. In the simplest useful model, a measurement is a weighted mixture of multiple excitations, filtered by frequency-dependent propagation paths. A spectral peak can therefore be physically real, temporally stable and still assigned to the wrong machine.

This matters because vibration standards distinguish the machine's own vibration from vibration transmitted from outside. ISO 20816-1 explicitly limits its evaluation criteria to vibration produced by the machine itself, not vibration transmitted to it from outside [7]. ISO 13373-1 emphasizes transducer placement, attachment, operating conditions and measurement procedures [6]. Neither statement implies that a general-purpose edge sensor can infer source ownership from a single mixed trace.

The research question is narrower than fault diagnosis: **Under what evidence conditions may a component recovered from a mixture be attributed to a named physical source?** A high reconstruction score, a low residual, or a distinctive harmonic pattern is not a sufficient answer. The proposed output is an evidence-bounded attribution decision that can also say unidentifiable.

Figure 1. Each sensor observes a mixture; spatial proximity does not assign source ownership.



The structural transfer paths are generally frequency-, placement- and operating-condition-dependent. The diagram is conceptual, not a representation of a measured SynapticSteel installation.

2. What separation does, and what attribution requires

2.1 Four different claims

- **Observation:** sensor j produced a valid sample/window with an identified device, axis, calibration, time basis and quality state.
- **Separation:** an algorithm generated candidate latent components that explain or partition some observed energy.
- **Attribution:** component k is linked to a particular physical machine through independently verified reference evidence and surviving alternative explanations.
- **Diagnosis:** a source-linked component supports a fault or condition inference under a separate validated diagnostic model. Attribution is necessary for some diagnoses, but is not sufficient.

These are not synonyms. Independent component analysis (ICA) identifies independent components under statistical assumptions but inherits permutation and scale ambiguities [1,2]. Second-order blind separation exploits distinct temporal covariance structures [3]. Nonnegative matrix factorization (NMF) can reveal

additive spectral parts, not automatically the machines that produced them [4]. A separator may be useful for detection while remaining unsuitable for machine-specific maintenance decisions.

2.2 Competing approaches and their failure modes

Approach	Useful when	Attribution limit
Single-sensor PSD, harmonics, order features	A speed reference or isolated source has distinctive orders; cheap and interpretable [11,12].	Shared orders, transfer resonances, leakage and changing speed can confuse identity.
PCA / ICA / SOBI	Multiple synchronized mixtures; favorable rank, independence or temporal structure [1-3].	Components are not physically labeled; correlated or near-identical motors weaken assumptions.
NMF / sparse time-frequency methods	Nonnegative magnitude mixtures and separable templates [4].	Magnitude additivity is an approximation; templates can be nonunique or drift.
Calibrated multichannel inverse model	Verified isolated references and stable transfer paths exist.	Conditioning, clock drift, unknown sources and calibration transfer dominate.
Supervised/deep enhancement	Enough representative labeled regimes exist; weak faults are masked [9,10].	Enhanced features can improve detection without establishing causal machine ownership.

3. A minimal model and two identifiability results

Let x(f,t) be an m-vector of synchronized sensor short-time Fourier transforms, s(f,t) a k-vector of source excitations, H(f,c) the m-by-k complex transfer matrix at operating context c, and n the noise. For a local linear, time-invariant approximation within a short operating regime:

```
x(f,t) = H(f,c) s(f,t) + n(f,t)
```

This is a modeling assumption, not a universal physical law. Changes in load, temperature, contact, mounting or rotational speed can invalidate the same H; nonlinear coupling and time-varying paths require richer models. The model is useful precisely because it makes an otherwise hidden identifiability problem explicit.

3.1 Result A: a perfect square fit cannot exclude an unknown source

Suppose m = k and H is nonsingular. For any observed x, the reconstruction s-hat = H-inverse x gives H s-hat = x (ignoring finite precision). Therefore the model residual is exactly zero regardless of whether the measured signal truly originated only from the k named machines. If an additional excitation u reaches sensors through transfer vector h4, then x = H s + h4 u. The estimated sources become s-hat = s + H-inverse h4 u. The unknown excitation is redistributed into named outputs. This is elementary linear algebra, not a newly proven theorem.

```
m = k, rank(H) = k => ||x - H H^-1 x|| = 0
```

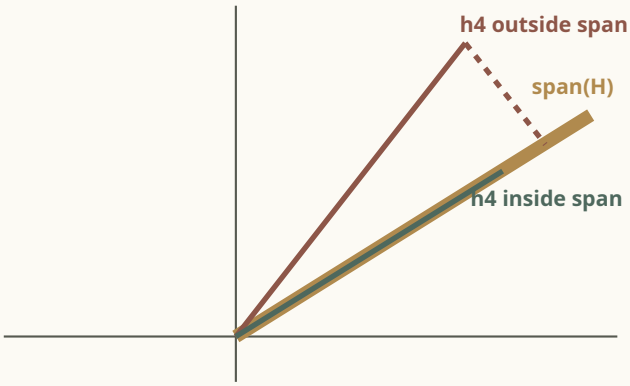
A square model with a small reconstruction residual can therefore be the least informative residual-based test for unknown sources. Even high correlation with an ideal synthetic mixture establishes only that the simulator matches its own assumptions.

3.2 Result B: an extra source inside the span is invisible to residual testing

For m greater than k, let P_H = H(H*H)^-1H* be the orthogonal projector onto the column space of full-column-rank H (where * is conjugate transpose). The least-squares residual is r = (I-P_H)x. If x = Hs + h4u + n, then:

```
r = (I - P_H) h4 u + (I - P_H) n
```

Only the component of h4 outside span(H) can appear in the ideal projected residual. If h4 lies in span(H), the unknown source is indistinguishable by this test even with additional sensors. More generally, detectability scales with the out-of-subspace transfer magnitude and the excitation amplitude relative to noise and model error. The phrase “a fourth source was not detected” must not become “there is no fourth source.”



What a residual can see

Out-of-span excitation:
nonzero projected residual
(if above noise and drift).

In-span excitation:
zero residual is possible,
even with a fourth motor.

Schematic: geometric relationship, not measured plant data.

Figure 2. The projection residual responds to the out-of-span part of an unmodeled source. An in-span source can be perfectly absorbed by the calibrated model.

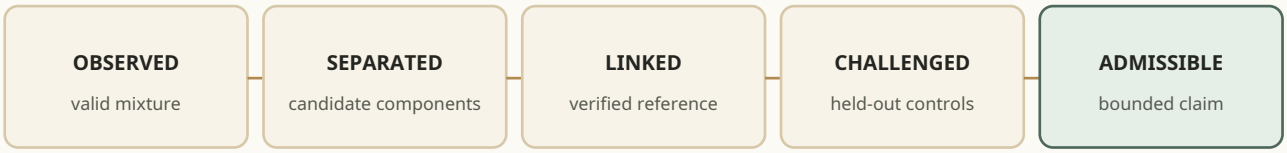
3.3 Conditioning is not a cosmetic diagnostic

When H is ill-conditioned, small measurement or calibration errors can cause large source-estimate errors. In a full-rank linear inverse, the relative perturbation bound is controlled by the condition number kappa(H) = sigma_max/sigma_min, subject to the usual assumptions. An example 3-by-3 matrix with two near-collinear columns has singular values approximately 1.883, 0.919 and 0.00485 (condition number about 388). A low residual then coexists with severe sensitivity to perturbations. The values are a constructed illustration, not observed machine measurements.

Rank and condition must be assessed by frequency and operating regime, not only once for the entire recording. Near-common motor orders, shared structural modes and nearly co-located sensor paths can collapse independent information in narrow bands. Counting nine accelerometer axes does not prove nine independent observations.

4. A Source Attribution Evidence Contract

The proposed **Source Attribution Evidence Contract (SAEC)** is an interface for limiting the strength of machine-specific claims. It does not mandate a particular separator. It demands that the separator expose the assumptions, provenance and failed tests needed for a downstream consumer to refuse overconfident attribution.



REFUSAL is an output, not a system error.

Unknown source, unverified clock, rank collapse, source swap, or reference drift must lower the claim level and preserve the original mixture and lineage.

Figure 3. The five-stage evidence ladder is a proposed claim-admission protocol. Refusal remains a first-class result at every transition.

4.1 Minimum evidence fields

Field group	Required receipt
Acquisition	board and axis ID; physical mount and orientation; sample rate and actual sample counters; calibration revision; window boundaries; clipping/gap/quality flags.
Clock and regime	common timebase evidence; offset and drift bounds; operating speed/load; stationarity window; timestamp uncertainty; clock-reacquisition epoch.
Transfer evidence	reference motor IDs; independent isolated-on/off proof; H revision, context and validity envelope; calibration dates; reference source hashes.
Numerical validity	per-band singular values/condition; rejected bands; noise floor; cross-validation and residual metrics; model family and hyperparameters.
Counterfactual evidence	held-out reference; source-swap challenge; fourth-source challenge; mount/load shift; unmodeled source alarm; false-attribution results.
Claim and authority	OBSERVED, SEPARATED, LINKED, CHALLENGED or ADMISSIBLE; explicit reason codes; confidence calibration if measured; no diagnosis/actuation authority by default.

4.2 Example schema fragment (proposal, not live API)

```
{"claim":"LINKED","machine_id":"motor-B", "clock_bound_us":250,"h_revision":"ref-03",
"min_sigma":0.018,"unknown_source_test":"INCONCLUSIVE",
"diagnosis_authority":false,"refusal_reason":null}
```

The numbers in this fragment are illustrative placeholders, not accepted engineering thresholds. A deployed contract would require typed units, schema version, signed or hashed evidence references, frequency-band validity, reference expiry and explicit policy mapping. The word “confidence” should be reserved for a calibrated probability or empirically validated score; fit fraction and spectral coherence are not confidence.

4.3 Refusal and graceful degradation

If sample clocks cannot be independently aligned, retain per-device vibration features but disable phase-sensitive demixing. If H loses rank, publish the affected band as unidentifiable. If an unknown source cannot be excluded, preserve an unlabeled component or an aggregate observation rather than forcing a named motor. If an isolation reference is stale after a sensor move, fall back to observational monitoring. These are useful degraded modes, not software failures.

5. Why familiar quality scores can mislead

5.1 Reconstruction and coherence

A normalized explained-energy score is useful for detecting gross model mismatch but cannot establish identity. With a square invertible H, it may be 1.0 by construction. Magnitude-squared coherence measures frequency-specific linear association between channels; shared drivers, structural coupling and correlated motor speeds can yield coherence without proving which motor is causal. GCC-PHAT estimates mixture-relative delay, not independent acquisition-clock synchronization or source ownership.

5.2 Permutation, scaling and domain shift

Blind separators can swap component order between windows, frequencies and operating regimes; scale is also ambiguous absent external calibration [1,2]. A model trained on one sensor mount or speed regime may transfer poorly to another. A source that changes speed can cross another source in the time-frequency plane. A classifier may retain apparent accuracy on an unrepresentative split while failing at the machine level when mounting, load or source composition changes.

5.3 Stronger signal is not stronger attribution

Modern deconvolution and learned separation can enhance weak repetitive fault signatures under interference. Ding and colleagues report experimental gains for a deep deconvolution method in rotating machinery [10], while Makienko and colleagues describe a vibration-separation/deconvolution method under stable operating conditions in a 2024 preprint [9]. These are useful demonstrations of enhancement or detection, not general evidence that a separated component is uniquely linked to a particular motor in a shared three-machine structural field. That additional identity claim requires its own controls.

6. Experimental design: make the hypothesis falsifiable

A credible study should test whether SAEC reduces wrong-machine attribution while preserving useful coverage, rather than merely improving a separation score. The primary hypothesis is that an evidence-gated method will have a lower rate of confident wrong-machine assignments than ungated inverse modeling on unseen operating regimes. A secondary hypothesis is that explicit abstention improves calibration at an acceptable loss of coverage. Both can fail.

6.1 Physical bench and ground truth

- Use three independently controlled motors on a shared frame or structure; instrument three spatially distinct STWIN-class acquisition points. Verify the number of actual boards and axes, their orientation, mounting, sensor calibration, sampling, and clock offset/drift before separation.
- Record isolated A/B/C motor runs across speed, load, temperature and mounting conditions. Independently record shaft-speed reference, power/on-off state and controlled intervention logs. Those references must not be derived solely from the same mixed vibration channels being scored.

- Create blinded mixtures with two/three motors, a fourth unknown excitation, one changed mount, shared orders, correlated speed schedules, clipping, missing windows, packet loss and forced clock drift. Include no-fault negative controls and hold out whole operating regimes and physical sessions, not merely adjacent time windows.
- Freeze the reference transfer matrix before held-out testing. Record failures to recover H after environmental changes, and test whether the method correctly abstains when conditions exceed the calibration envelope.

6.2 Baselines, ablations and outcomes

Test arm	Role
A. nearest-sensor PSD/order features	Low-cost operational baseline; assess simple physical proximity and harmonic heuristics.
B. ICA/SOBI or NMF	Blind/semiblind separation baseline; compare recovered components and label instability.
C. calibrated inverse H	Reference-aware source reconstruction without the proposed admission/refusal gates.
D. C + SAEC	Full claim gating; refusal and provenance receipts.
E. D without rank / clock / fourth-source gate	Ablations to test which gates actually reduce false attribution.

Primary endpoint: wrong-machine assignment rate among accepted named-source claims, with an explicit denominator and 95% interval across independent sessions. **Paired endpoint:** coverage (fraction of eligible windows receiving a named claim) to prevent a trivial all-abstain “success.” Additional endpoints: per-source F1/balanced accuracy, out-of-distribution rejection, calibration error for any claimed probability, phase/clock robustness, runtime/memory and sensor failure behavior. Report a risk-coverage curve, not one cherry-picked operating point.

Pre-registration: select tolerable wrong-assignment risk and minimum useful coverage before looking at held-out results. Thresholds are application- and consequence-specific; this paper does not invent universal industrial acceptance numbers. Compare paired results across session-level splits and preserve negative results. A failure to beat simple order features is scientifically meaningful.

6.3 Deliberate adversarial tests

- **In-span fourth source:** construct or identify an extra excitation whose transfer pattern is nearly representable by H; show whether a low residual incorrectly authorizes A/B/C labels.
- **Near-collinear paths:** move sensors or select structural resonances to reduce sigma_min; measure label instability and gate behavior under injected calibration noise.
- **Reference spoof by regime change:** alter load or attachment after calibration without updating the reference; test refusal and recalibration triggers.
- **Clock shift:** introduce bounded drift and jitter with an independently measured reference; distinguish sample-time error from acoustic propagation delay.
- **Source exchange:** swap motor control schedules and labels while keeping overlapping spectral orders; ensure the algorithm follows verified physical IDs rather than component index.

7. Experimental scope and implementation boundary

This paper is a research proposal and evaluation protocol. Private operational checkpoints are excluded from this public edition. No implementation, deployment, field validation, or measured performance benefit is established by the proposal. A prototype should first be evaluated with recorded or simulated inputs, explicit baselines, failure cases, and independently measured outcomes.

8. Counterarguments and practical alternatives

8.1 “Put one accelerometer on each machine”

Often the correct engineering answer is better mounting and simpler per-machine sensing. If reliable direct attachment isolates each machine sufficiently, SAEC adds overhead with little value. But shared foundations, inaccessible equipment, acoustic channels and sparse deployments still create mixture problems. The proposed contract should be justified only when the ambiguity has consequential cost and cannot be removed cheaply by physical instrumentation.

8.2 “Deep models learn the source identity”

A sufficiently varied, independently labeled dataset may support a robust classifier without an explicit transfer matrix. That is a legitimate competitor, especially under nonlinear dynamics. Its burden shifts to representative training, out-of-distribution detection and empirical calibration. SAEC can wrap learned models, but must not pretend rank tests alone validate a nonlinear representation.

8.3 “Abstention makes the product less useful”

Yes, refusal reduces coverage. A system that never attributes anything is perfectly safe by one narrow metric and operationally useless. That is why wrong-attribution risk and accepted-claim coverage must be reported together. The architecture is justified only if its risk-coverage tradeoff improves over simpler baselines at a useful operating point.

8.4 “The linear model is too simple”

Also true outside locally stable conditions. Structural resonances, time-varying RPM, nonlinear contacts and temperature-dependent paths complicate H. The point of a deliberately explicit model is to reveal where claims are unidentifiable, not to imply all factories obey it. More complex physics-informed or learned models can be substituted, but they still require independent source references and counterfactual challenge.

9. Implementation sequence and limitations

A minimal research implementation would first collect trustworthy raw synchronized waveforms and independent source-state logs, then freeze isolated references and their validity envelopes, then compute per-band rank/conditioning and a conservative demixer, then add fourth-source/mount-change challenges, then report risk-coverage. A production adaptation would require governed sensor admission, bounded compute and retention, rollback, and clear separation between model inference and operator/maintenance authority.

Limitations: The mathematical examples assume linear additive mixing within a stationary analysis window; frequency-domain projection does not solve nonlinear or convolutive identifiability in general. Unknown-source tests cannot rule out all adversarial excitations. Reference isolation may be operationally expensive or unsafe. Ground-truth interventions must be planned by qualified operators, not automated by

the research software. A three-motor bench is not representative of arbitrary industrial plants. Neither the proposed receipt schema nor a numeric threshold is validated by this paper. No original physical experimental results are presented.

Research outlook: Study source-transfer uncertainty sets, Bayesian model selection over unknown sources, sensor-placement optimization for singular-value margin, uncertainty propagation through downstream temporal models, and human-verifiable counterfactual explanations. These are hypotheses and proposed studies, not product claims.

10. Conclusion

A separated vibration component is not yet a machine. When a full-rank square model can fit any mixture and an extra source may lie inside the modeled transfer subspace, reconstruction quality cannot establish source identity. A defensible system must bind machine attribution to independent references, clock and geometry evidence, conditioning, regime validity, adversarial alternatives and the option to refuse. The proposed SAEC makes that boundary explicit and testable. Its merit remains an empirical question: it should be judged by fewer wrong-machine assignments at useful coverage, not by the elegance of the separator or the smoothness of its plots.

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Related SynapticSteel Technical Papers

This paper is technically adjacent to **STP-003, The Timestamp Is Not the Event** (event-time uncertainty), **STP-005, Trust Is Not Agreement** (independent evidence versus consensus), **STP-006, The Probe Is Part of the Process** (observation impact), and **STP-007, A Reconnection Is Not a Recovery** (sensor reacquisition). These are series relationships, not claims that all related papers have been published. No unverified related DOI is asserted.

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Source boundary: Primary literature and standards form the background for this proposal. The elementary analytical examples are not measurements from industrial equipment. No diagnostic performance improvement is claimed.

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